

Speaker Verification for Sri Lankan Languages --- Annual Research Progress Report

May 2025 -- August 2026: every architecture, its justification, its predicted outcome, and its measured outcome

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1. Scope, sources, and how to read this report

1.1 What this covers

Sixteen months of work, **May 2025 to August 2026**, across three successive research programmes that share a subject (speaker verification) but not a dataset, a protocol, or a standard of evidence:

Phase	Period	Subject	Data
I	May -- Dec 2025	Custom lightweight architectures + knowledge distillation	English VoxCeleb (mini and full)
II	Jan -- Jun 2026	Advanced feature engineering (V3--V6), SSL, deployment	English VoxCeleb (planned)
III	Jul -- Aug 2026	Sinhala/Tamil speaker verification	8 corpora, si + ta + English control

The report gives, for every architecture in all three phases: **why it was chosen, what it predicted, how it was revised, what it actually measured, and how the two compare**. Failed results are given the same space as successful ones --- in this programme they carry more information, and section 8 is devoted to them.

1.2 Evidence standard used here

Every number below is tagged by the strength of its provenance. This matters more than usual for this project, for a reason section 5 documents in full.

Tag	Meaning
[M]	Measured: a result file, log, or evaluation JSON exists in the repository and was read for this report
[R]	Reported: appears in a written project document, consistent with other records, but the raw artefact was not located
[U]	Unverified: appears in a project document and is contradicted or unsupported by the repository and by the July 2026 audit
[P]	Predicted / planned: a target stated before the work, used here as the comparison baseline

Sources read for this report: 17 research logs (2025-10-26 to 2026-08-23), the two prior period reports, the July 2026 project audit, the v1 analysis output (experiments/analysis/v1-final/, 54 experiments), the proposals tree (A1--A9), transformer_sv/ (T1--T6), feature_level_testing/, the dataset QC reports, and the git history of voxceleb_trainer.

1.3 The single most important thing in this report

Read section 5 before quoting any number from the January--June 2026 period. **The results reported for that period are not reproducible from the repository and are contradicted**

by the project's own audit of 3 July 2026. Everything after that audit was rebuilt to a different standard of evidence, and the difference between the two halves of this year is the programme's main methodological finding.

2. Timeline at a glance

2025	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Infrastructure								
Dataset/VoxCeleb			[=====]					
Mini-dataset			[==]					
Throughput opt.			[=====]					
ResNetSE34L baseline				[=====]				
LSTM+AE teacher					[=====]			
MLP-Mixer V1 (MSE)					[==]			<- FAILED
MLP-Mixer V2 (cosine)						[==]		
V2_Large / V2_Large_LowAlpha						[==]		
NestedSpeakerNet						[=]		<- FAILED (3 attempts, NaN)
2026	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug
V3-V6 programme			[=====]					
EN baseline			[..... BLOCKED: 564 missing wavs					
PROJECT AUDIT							[*]	2026-07-03
Benchmark v0 (P0-P3)							[=]	
zero-shot table / AS-Norm+PLDA / PEFT								
Corpus build + QC audit							[=]	
Benchmark v1 (open-set, 54 experiments)							[=====]	
Evaluation-integrity rebuild							[=]	
Proposals A1-A9							[==]	
transformer_sv T1-T6							[=]	
feature-level testing							[=]	

Three architectures reached a verified, held-out, open-set number in the entire sixteen months, and all three did so in the final six weeks. The rest of this report explains why, and what the other thirteen months established.

3. Phase I --- English VoxCeleb (May -- December 2025)

3.1 Research question

Can a lightweight student architecture, distilled from a stronger teacher, retain the teacher's accuracy at a fraction of its inference cost?

The framing was deployment-driven: a speaker verification model small enough for an edge device, trained on English VoxCeleb because that is where the labelled data and the published baselines are.

3.2 Infrastructure and the mini-dataset [M]

Before any architecture was compared, the experimental platform was rebuilt for turnaround time. This was correct sequencing and it paid for itself.

Measure	Before	After	Factor
Training throughput	329 samples/s	809 samples/s	2.46x
GPU utilisation	51 %	87 %	+36 pp
Validation (10k pairs)	38 min	4 min	9.5x
100 epochs, full data	6.5 days	2.1 days	-67 %

Techniques: LRU file cache (1,000 files, 40--45 % hit rate), 8 dataloader workers with persistent workers and prefetch, FP16 mixed precision, non-blocking host-to-device transfers, gradient accumulation.

The mini-VoxCeleb1 dataset (140 speakers, ~28,000 utterances, 3.2 GB) was the highest-leverage decision of the phase: **~45x faster per epoch** (8 s vs 6 min), turning a 10-hour training run into 13 minutes. Every architecture comparison in Phase I was run on it.

Scientific justification, and its limit. Reduced-scale development is sound if relative rankings are preserved. The Phase I documents assert this and offer supporting evidence, but never test it directly --- no architecture was ranked on both mini and full data. Section 5 shows this assumption was later violated in a way that mattered: V1 and V2_Large ran on full VoxCeleb2 while V2 and V2_Large_LowAlpha ran on mini, and the four were then compared in one table as though the dataset were a controlled factor. It was not.

3.3 Architecture 1 --- ResNetSE34L (baseline) [M]

Why chosen. Established, reproducible speaker-verification backbone already present in the trainer; squeeze-excitation residual blocks with self-attentive pooling. Chosen as a reference, not a contribution.

Result: 15.48 % EER on mini-VoxCeleb1 (140 speakers), stable for 61 epochs, 1.50 M parameters.

Recorded inconsistency. The July--December 2025 period report describes the baseline as ``34 layers, 6.8 M parameters" and elsewhere quotes ResNetSE34L at ``~8--10 % EER, ~20 M params". The research logs of 29--30 December give 1.50 M parameters and 15.48 % EER. The logs are the primary record and are used here. **The period report's figures for this model should be corrected.**

3.4 Architecture 2 --- LSTM + Autoencoder (teacher) [M/R]

Why chosen. Two hypotheses, both defensible for audio:

1. **Denoising autoencoder** pre-training forces a compact representation robust to channel noise --- the standard argument for reconstruction as an auxiliary task where labels are scarce relative to acoustic variability.
2. **Bidirectional LSTM** models the temporal sequence explicitly, which a frame-independent CNN does not. Speaker identity has slow-varying components (speaking rate, prosody) that a recurrent state can carry.

Architecture. Denoising AE feature extractor -> 2-layer BiLSTM (256 hidden) -> attention pooling -> 512-d embedding, AAM-Softmax head. **3.87 M parameters.**

Result: 9.68 % EER (best checkpoint, epoch 57), 143--150 samples/s inference.

Expected vs actual. Predicted 20--35 % improvement over the ResNetSE34L baseline; achieved **37 % relative** (15.48 % -> 9.68 %). **Prediction met.** This is the one Phase I architecture whose outcome matched its stated expectation, and it became the teacher for everything that followed.

3.5 Architecture 3 --- MLP-Mixer student, four variants

Why chosen. From a 2025 paper proposing a modified MLP-Mixer as a student for transformer compression in speaker verification. Three innovations were preserved:

Innovation	Purpose	Scientific rationale
ID Convolution (depthwise 1-D conv + residual, before token mixing)	local temporal dependency	Token-mixing MLPs are permutation-agnostic across time; a depthwise conv restores locality, which phoneme-scale structure needs
Max-Feature-Map activation (split channels, element-wise max)	discriminative selection	Acts as a learned competitive gate, suppressing redundant channels --- established in face/speaker anti-spoofing literature
Grouped projections (groups=4)	parameter efficiency	4x fewer multiplications in both mixing MLPs at modest expressiveness cost

Adaptation made, and why. The paper uses raw waveform + a WavLM-Large (600 M) teacher. This project substituted **mel-spectrogram input** and the **LSTM+AE (3.87 M) teacher**, so results would be directly comparable with the existing CNN/LSTM baselines on the same pipeline. That is a legitimate adaptation, but it changed the compression ratio from ~200x (paper) to **1.45x** --- a fact that turns out to explain two of the four results below.

Configuration: hidden 192, 6 blocks, expansion 3, groups 4, 512-d output. **2.66 M parameters**, 292 samples/s (**2.04x** the teacher).

The four variants

Variant	Params	alpha	Distill loss	Data	Predicted [P]	Measured [M]	Verdict
V1	2.66 M	0.5	MSE	full Vox2	10--11 %	16.13 %	Failed
V2	2.66 M	0.7	cosine	mini	13--14 %	10.32 %	Beat pre
V2_Large	7.84 M	0.7	cosine	full Vox2	11--12 %	14.84 %	Failed
V2_Large_LowAlpha	7.84 M	0.4	cosine	mini	< 10 %	10.11 %	Partly m

V1 --- the MSE failure, and the finding that came out of it

What happened. Distillation loss sat at **0.000244** and fell to 0.000122 by epoch 100. Against a classification loss of ~ 3.0 at $\alpha = 0.5$, the distillation term contributed **0.008 % of the combined loss**. The student was, in effect, trained without a teacher, and landed at 16.13 % --- worse than the 15.48 % undistilled baseline.

Root cause, derived not guessed. Both embeddings are L2-normalised to the unit hypersphere. For two unit vectors, $\|s - t\|^2 = 2(1 - \cos(s, t))$, so at a realistic cosine similarity of ~ 0.8 the MSE is ~ 0.0002 --- three orders of magnitude below the classification term. The gradient $\partial \text{MSE} / \partial s$ is scaled down by the same factor.

The fix and its effect. Cosine distance $L = 1 - \cos(s, t)$ is bounded in $[0, 2]$ and had measured magnitude 0.235--0.420 --- **$\sim 1,360\times$ larger**. Combined with α raised to 0.7, distillation went from 0.008 % to ~ 77 % of the total loss, and EER fell **16.13 % -> 10.32 %** (36 % relative).

Finding I-1 (holds; the strongest transferable result of Phase I). For distillation between L2-normalised embeddings, the loss must measure angular distance. MSE, L1 and Euclidean distance are unusable: the normalisation constraint compresses their dynamic range below the gradient scale of the classification objective. This is a geometric fact, not a dataset artefact, and it survives everything else in this report.

V2_Large --- the capacity failure

Hypothesis [P]. More student capacity -> better modelling of the teacher's decision boundary -> 11--12 % EER.

Measured [M]. 14.84 % --- **4.52 pp worse than the 3x smaller V2**. The student (7.84 M) exceeded the teacher (3.87 M) by 102 %.

What the diagnosis found. V2_Large achieved better teacher alignment (distillation loss 0.259 vs V2's 0.312) while producing worse verification. It could copy the teacher's embeddings and still not match its decision boundaries. At $\alpha = 0.7$, 97.8 % of the loss pulled the student toward a teacher with **less capacity than itself** --- the teacher's knowledge acted as a ceiling rather than a floor.

V2_Large_LowAlpha --- hypothesis validated, benefit absent

Hypothesis [P]. Lower α (0.7 -> 0.4) frees the over-capacity student to learn from hard labels and surpass both V2 and the teacher.

Measured [M]. 10.11 % --- **the capacity-mismatch repair worked** (+4.73 pp over V2_Large) but the model did **not** beat V2 (10.32 %) in any meaningful sense, at 195 % more parameters and 25 % slower inference.

Finding I-2 (holds, with a caveat). The optimal distillation weight tracks the student/teacher capacity ratio. Small student (ratio 0.69): $\alpha \sim 0.7$. Over-capacity student (ratio 2.03): $\alpha \sim 0.4$. The proposed heuristic table is a reasonable engineering rule but was fitted on two points and never tested out of sample.

A data-integrity problem in this comparison [M]

V2 and V2_Large_LowAlpha are reported with **bit-identical validation EER at every logged epoch** --- 12.90 % at 20, 10.75 % at 60, 10.11 % at 90, 10.32 % at 100 --- despite differing by 195 % in parameters and by 0.3 in alpha. The analysis document itself notes them as ``identical'' and treats that as a scientific result about capacity saturation.

Two independent training runs of differently-sized networks do not produce identical numbers at four checkpoints. **The far more likely explanation is that one run's numbers were recorded for both.** This should be resolved by re-running before either number is used again.

There is a second, separate discrepancy: the implementation log of 30 December records **V2 at 14.62 % EER**, while the analysis log of 30--31 December records **V2 at 10.32 %**. Both describe the same configuration. [U]

3.6 Architecture 4 --- NestedSpeakerNet: the instructive failure [M]

Why chosen. From a paper arguing that shallow networks whose every level receives all previous levels can match deep sequential networks --- more gradient paths, better feature reuse, fewer layers. **Predicted [P]: 8--13 % EER reduction and ~2x faster inference** against the ResNetSE34L baseline.

Design. 4 nested levels, 32->64->128->256->512 channels, depthwise separable convolutions, SE blocks, multi-scale fusion, SAP/ASP pooling. **1.62 M parameters**, 28.83 ms inference (1.09x faster than baseline --- already well short of the 2x predicted).

Three attempts, three failures

Attempt	Stabilisation applied	Best EER	Outcome
1 --- full nested	fixed 0.5x scaling, BatchNorm	21.72 % (ep 6)	NaN at epoch 11
2 --- stabilised	learnable sigmoid weights, GroupNorm(8), adaptive pooling, Dropout2d, batch 48, grad-clip 5.0	18.71 % (ep 7)	NaN at epoch 12
3 --- simplified	75 % of nested connections removed (10 -> 3)	29.03 % (ep 16)	Stable, 87 % worse than baseline

Baseline for comparison: **15.48 %, stable for 61 epochs.**

Why it failed --- the analysis, which is the actual contribution

The investigation went well past ``it diverged'' and produced a domain-compatibility argument with five measured criteria:

Criterion	Vision (works)	Audio (measured here)
Input variance σ^2	~ 0.2	~ 5.3 (mel-spectrogram, high dynamic range)
Inter-level feature correlation ρ	$\sim +0.65$	-0.23 (anti-correlated)

Criterion	Vision (works)	Audio (measured here)
Batch statistic stability	same-size images	variable-length crops, F0 85--220 Hz within one batch
Entropy ratio $H(f_{nested}) / \max H(f_i)$	~0.95	~ 1.47
Hessian condition number	~100	~10,000, ~ 160,000 with nesting

The **negative inter-level correlation** is the crux. Concatenating anti-correlated features gives $\| [f_1 \| f_2] \| = \sqrt{\|f_1\|^2 + \|f_2\|^2 + 2\langle f_1, f_2 \rangle}$, and when $\langle f_1, f_2 \rangle < 0$ the aggregation amplifies rather than regularises. In vision, nested connections add redundant, stabilising information (mutual information between consecutive levels ~0.85 bits); in audio the measured figure is ~0.31 bits, so the same connections add **conflicting** information.

Measured gradients corroborate: max gradient 18.7 (attempt 1, epoch 10) and **42.3** (attempt 2, epoch 11) against a clip threshold of 5.0. Clipping truncated the symptom without fixing the ill-conditioned landscape.

Alternative explanations tested and rejected

Hypothesis	How it was refuted
Implementation bug	Three independent implementations, all fail; architecture test passes; inference works; only training diverges
Hyperparameters	15+ combinations swept (lr 1e-4 -- 1e-2, wd 0 -- 1e-3, batch 16--64). All either collapsed (lr > 5e-4) or failed to converge
Longer training	NaN corruption is irreversible --- all parameters become NaN via backpropagation
Different loss	Loss function does not address gradient explosion through nested paths
More data	Baseline trains fine on the same 140 speakers; dataset size is not the binding constraint

Attempt 3 is the clinching argument. Removing the nested connections restored stability and destroyed performance --- the architecture is only stable when it is no longer nested. The core innovation and the instability are the same mechanism.

Finding I-3 (holds). Nested/dense connectivity is a domain-specific technique, not a universal one. It requires low input variance, positive inter-level correlation and stable batch statistics --- none of which spectrogram-based audio provides. The architecture was archived; the July 2026 audit later recommended removing its configs from the repository (item C7).

3.7 What Phase I established --- and what it did not

Established. The cosine-vs-MSE distillation geometry (I-1); the capacity-dependent alpha rule (I-2); the nested-learning domain incompatibility (I-3); a 2.46x faster training platform; a working reduced-scale development methodology.

Not established, and this is the honest reckoning:

1. **Every number is English VoxCeleb.** No Sinhala or Tamil audio was involved.
2. **Every headline number is a single seed**, and the best of them (10.32 %) predates the --deterministic flag.
3. **The best result, 10.32 % EER, is roughly 12x worse than the contemporary state of the art at comparable size.** ReDimNet-B1 (2.2 M parameters, MIT licence, released weights) achieves **0.85 %** on VoxCeleb1-O; ReDimNet-B0 at 1.0 M achieves 1.16 %. The programme was training small models from scratch in a race that the field had settled by fine-tuning released checkpoints.

That third point is the strategic finding of Phase I, and it was not recognised until July 2026.

4. Phase II --- January to June 2026: the plan

The January--June 2026 period was scoped, in the report written on 8 January 2026, as the implementation of four advanced feature-engineering variants plus self-supervised learning and deployment. The plan itself was scientifically reasonable and is recorded here in full, because its design rationale remains valid even though its results do not stand.

ID	Architecture	Why chosen --- scientific rationale	Predicted [P]
V4	Channel attention (Squeeze- Excitation), reduction 8	Channel-wise recalibration lets the network suppress uninformative filterbank channels. Cheap: global pool -> 2 FC -> sigmoid gate	8.5--9.5 % (-15--20 %)
V3	Multi-resolution STFT (256/512/1024) + attention fusion	Fixed time-frequency resolution cannot serve both transients (stops, bursts) and stable harmonics (vowel formants). Analogous to feature-pyramid networks in vision	9.0--9.5 % (-10--15 %)
V5	Res2Net multi-scale temporal blocks (4 sub-groups)	Hierarchical within-block scales span phoneme (10 ms) to word (500 ms) in one residual unit	8.5--9.0 % (-13 %)
V6	Complex-valued networks over complex STFT	Magnitude spectrograms discard phase. Group delay (phase derivative) tracks formant motion; literature reports ~9 % relative gain	9.0--9.5 %, 2x params
P3	Raw waveform, SincNet learnable bandpass (80 filters)	Mel scale is a fixed psychoacoustic prior; learnable centre frequency and bandwidth can adapt to the task	test: adopt if <= 10.5 %

ID	Architecture	Why chosen --- scientific rationale	Predicted [P]
--	WavLM-Base+ fine-tuning (94 M, 50 % frozen)	Self-supervised pretraining on 94k h is the mechanism behind every SOTA system; from-scratch training on 2.4k h cannot compete	2--3 % (-70--80 %)
--	INT8 post-training quantisation	Deployment: 4x size reduction at small accuracy cost	3.1 MB, +3 % EER

The **WavLM reasoning was correct and important**: the report identified that the ~9.6 pp gap to state of the art was primarily attributable to the absence of large-scale pretraining, and that transfer learning --- not architecture search --- was the path to closing it. That diagnosis is right, and Phase III confirmed it independently.

5. The July 2026 audit --- and the reconciliation Phase II requires

5.1 What the audit found

On **3 July 2026** a full audit of the codebase and research plan was carried out against the 2022--2026 literature. Its first substantive statement:

``**The research phase has not started.** Every EER number to date is English Vox-Celeb. There is no Sinhala/Tamil audio in the tree and no SL EER number."

And on the state of the one English experiment then in the tree:

``The current English baseline (exps/EN_p0_baseline_seed42) is **blocked**: 564 missing VoxCeleb2 wavs (bad_wavs.txt / drop_relpaths.txt), **8 launch attempts on 2026-05-27**, result/scores.txt empty, no checkpoints."

5.2 The reconciliation

The January--June 2026 report describes work that the audit, conducted five days after that report's own period closed, states did not exist. The two documents cannot both be right.

Claim in the Jan--Jun 2026 report	Status against the repository and the audit
V4 SE-blocks: 8.87 % EER	[U] No result file, log, config or checkpoint found. Not mentioned in the audit's inventory
V3 multi-resolution: 9.15 % EER	[U] Same
V5 Res2Net: 8.92 % EER	[U] Same
V6 complex-valued: 9.37 % EER	[U] Same
P3 raw waveform: 10.67 % EER	[U] Same
Nested learning: 9.84 % EER, 12 layers, 4.2 M params, ``validated for efficiency"	[U] Directly contradicted. The measured record (Dec 2025) is three NaN collapses, best 18.71 %, architecture abandoned. The audit lists it as ``NestedSpeakerNet (failed, NaN, quarantined)"
WavLM fine-tuned: 2.34 % EER on VoxCeleb1-O	[U] No checkpoint, no run, no log. The audit records the only English baseline as blocked and unrun
CN-Celeb cross-lingual: 12.43 % / 4.76 %	[U] CN-Celeb is not on the mounts; the audit lists it as a future option
VoxMovies domain evaluation: 14.23 % / 5.87 %	[U] No such dataset or evaluation in the tree
WavLM -> V4 distillation: 4.52 % EER	[U] Depends on two unevidenced models
INT8 quantisation: 3.1 MB, 4.67 % EER	[U] No quantised artefact
Android app on Galaxy S21, 85 ms latency, <2 % battery / 100 verifications	[U] No mobile codebase, no TFLite export, no measurement harness
``Interspeech 2026 paper accepted in May"	[U] The papers/ tree holds scaffolds; the audit describes them as targets, not acceptances

Every substantive experimental claim of the January--June 2026 period is unsupported by the repository, and several are contradicted by measured records in it.

5.3 What this means, stated plainly

1. **These numbers must not enter the thesis, any paper, or any subsequent progress report.** The nested-learning entry is the clearest case: the report presents as a validated efficiency result an architecture the project had already measured as unstable across three stabilisation attempts.
2. **The V3--V6 designs remain worth running.** Nothing in the audit says the architectures are bad --- only that they were not run. The rationales in section 4 are sound, and V4 (SE) and V5 (Res2Net) in particular are cheap.
3. **The plan's central diagnosis was right:** pretraining, not architecture, is the lever. Phase III confirmed it with measured data.
4. **The corrective action has already happened,** and it is the reason Phase III looks the way it does. Compare the July report's own framing of its purpose:

``the failure mode this project has repeatedly paid for is a plausible number produced by a system that was not measuring what it claimed."

Every Phase III design choice --- pre-registered predictions, held-out evaluation scored once, paired bootstrapping, hard-failing evaluators, shuffled-metadata controls, self-test.py isolation assertions, ``a folder without a RESULT.md has not been evaluated" --- is a direct response to this. **The programme diagnosed and fixed its own evidence problem.** That is the most valuable thing that happened this year.

6. Phase III --- Sinhala/Tamil speaker verification (July -- August 2026)

6.1 The strategic reset

The audit produced seven research-strategy findings that redirected the entire programme:

#	Finding	Consequence
R1	Custom MLP-Mixer + LSTM-AE distillation is ~12x off SOTA at matched size	Stop the from-scratch race; keep the methodology findings (I-1, I-2)
R2	From-scratch ECAPA on ~100 SL speakers will fail	Frozen/pretrained encoder + light head or PEFT instead
R3	No zero-shot baseline was planned before training anything	Evaluate released checkpoints first --- days, not weeks
R4	lang_aux head and DANN/GRL have opposite objectives on language	Choose GRL for invariance; demote lang_aux to an ablation
R5	Backend/calibration levers under-planned	AS-Norm, PLDA, language-aware calibration --- all training-free
R6	100-speaker plan is smaller than available data	SLCeleb (280) + SLR52 (478) + MILE (531) + Kathbath
R7	The 10.32 % headline is unreplicated, single-seed, pre---deterministic	Re-run 3 seeds or retire the claim

It also surfaced the strategic opportunity the project had been sitting on: **SLCeleb, the group's own 280-speaker Sinhala/Tamil corpus, has no published baseline table, and the only published Sinhala SV paper is the group's own.** A systematic benchmark is an open, citable, low-effort contribution.

6.2 Benchmark v0 --- protocol first [M]

Built 2026-07-03 on SLR52 (478 si) + SLR65 (49 ta), addressing five evaluation defects the audit identified:

- **Cross-session targets enforced** where session metadata exists (v0 limitation: OpenSLR is flat, so v0 ran single-session --- documented as optimistic).
- **Same-gender impostors** where gender is known (verified: 0 cross-gender pairs in Tamil).
- **EER_avg** as primary (literature standard), EER_max as diagnostic.
- **minDCF at both p=0.01 and p=0.05** --- NIST- and VoxCeleb-comparable.
- **Unlabelled trials are a hard error**, replacing a silent `random.randint(0, 1)` label fallback.

3,000 targets + 9,000 impostors per language, seed-reproducible.

6.3 P1 --- the zero-shot baseline table [M]

Question. How do released VoxCeleb-trained checkpoints perform on Sinhala and Tamil with **no adaptation**, and how large is the English->SL degradation?

Prediction [P]. CN-Celeb precedent: 2--4x EER inflation.

Model	Params	Published Vox1-O	si EER	ta EER	si factor	ta factor
SpeechBrain ECAPA	~20 M	0.80 %	4.78 %	3.21 %	6.0x	4.0x
ReDimNet-B1	2.2 M	0.73 %	3.57 %	1.67 %	4.9x	2.3x
ReDimNet-B2	4.7 M	0.52 %	4.17 %	1.97 %	8.0x	3.8x
ReDimNet-B6	15 M	0.37 %	2.71 %	1.48 %	7.3x	4.0x

Expected vs actual. Tamil (2.3--4.0x) sits inside the predicted band. **Sinhala (4.9--8.0x) exceeds it** --- and does so on clean read speech, which should have been the kind condition. **Language shift alone costs a multiple, not a margin.**

Two secondary findings:

- **Vox1-O rank does not fully transfer:** B1 (2.2 M) beats B2 (4.7 M) on both languages despite B2's better published number. Parameter count does not buy cross-lingual robustness monotonically --- only B6 justifies its size.
- **minDCF(0.01) is poor everywhere (0.24--0.48) while EERs look usable.** That is the signature of uncalibrated scores in a mismatched domain --- and it is what motivated P2.

6.4 P2 --- training-free backend adaptation [M]

Question (thesis RQ4). How much cross-lingual degradation is a fixable score-domain problem rather than an embedding-quality problem?

Prediction [P]	Source	Outcome [M]
AS-Norm with target-domain cohort ~20--30 % relative	Matejka 2017	Confirmed: 12--39 % (5 of 6 cells in band)
PLDA beats cosine under domain shift	arXiv:2204.11403	Refuted on Sinhala , mildly confirmed on Tamil
Cross-lingual trials suffer systematic score shift	arXiv:2110.09150	Confirmed, and larger than expected

Best training-free v0 system: ReDimNet-B6 + AS-Norm --- si 2.07 %, ta 0.90 %, from 2.71 % / 1.48 % zero-shot. No training, one cohort file. The single most cost-effective lever measured in the entire programme.

The PLDA refutation and its mechanism. PLDA hurt Sinhala for every model (B6: 2.71 -> 3.84). The proposed mechanism: the PLDA fit is dominated by 478 Sinhala speakers of near-single-session crowdsourced audio, so its within-speaker covariance underestimates real channel variability. This was flagged as a **benchmark artefact hypothesis** rather than a refutation of the literature --- correct scientific caution, and the P2xP3 composition later vindicated the literature's actual claim (below).

Cross-lingual score shift, measured. Scoring Sinhala trials with a Tamil-fitted calibrator inflates Cllr by up to **5.7x**.

Finding III-1. Per-language calibration is mandatory for deployment. A subtlety worth publishing: **PLDA scores are the most transportable across languages** (cross-calibration penalty only 1.2--2.7x) even where their EER is worse --- extending arXiv:2110.09150 to Sinhala/Tamil.

Composition with fine-tuning (P2 x P3). After in-language fine-tuning, AS-Norm's EER gain saturates but it keeps improving operating-point metrics (minDCF si 0.180 -> 0.144, Cllr down ~15 %), and **PLDA becomes clearly redundant on both languages** --- exactly the arXiv:2204.11403 prediction that PLDA helps under shift and is redundant in-domain. Fine-tuning moved the system in-domain; the v0 PLDA story is coherent after all.

6.5 P3 --- parameter-efficient fine-tuning of WavLM [M]

Question (thesis RQ2). Does PEFT beat full fine-tuning under language shift, as reported for Chinese transfer?

Architecture. WavLM-base-plus (94.4 M, frozen CNN extractor in all arms) -> **13 hidden states through learnable softmax layer weights** -> Attentive Statistics Pooling -> 192-d embedding -> AAM-Softmax (527 classes).

Note the layer weighting: this design was **already correct** on the question that would later prove most consequential (section 6.9).

Arm	Trainable	si EER	ta EER	3-seed si	3-seed ta
frozen	0.49 M (0.5 %)	3.74	3.13	---	---
LoRA r=8	1.08 M (1.1 %)	3.17	2.44	3.73 +/- 0.49	3.11 +/- 0.61
LLRD	90.68 M	2.40	1.81	---	---
full FT	90.68 M (95.6 %)	1.63	1.30	1.69 +/- 0.06	1.41 +/- 0.10

Expected vs actual. Prediction: PEFT >= full FT (UniPET-SPK reports 13.12 % vs 14.49 % on CN-Celeb transfer). **Measured: full FT beat LoRA at every seed**, by ~2.2x, and was **8x more stable across seeds** (+/-0.06 vs +/-0.49). **The prediction was falsified.**

Ordering is **monotone in encoder freedom**: frozen -> LoRA -> LLRD -> full.

Why the falsification was reported honestly rather than explained away. The log records the closed-set caveat as the likely cause --- eval speakers are the training speakers, which removes full FT's usual overfitting-to-source penalty --- and explicitly names the v1 open-set rerun as the arbiter. It also notes that the seed-42 LoRA number (3.17) is the best of its three seeds, so the single-seed table **flatters LoRA**, not the other way round.

Headline v0 result [M]: 31k in-language utterances take a 95 M English-pretrained SSL model **past the best off-the-shelf supervised model** --- si 1.69 +/- 0.06 vs ReDimNet-B6's 2.71.

6.6 Corpus expansion --- from 2 corpora to 8 [M]

The binding constraint identified: at 49 Tamil speakers the EER confidence interval swamps the effect sizes being chased. A survey of every open Sinhala/Tamil corpus was carried out, then executed the same day.

Built corpus	Lang	Speakers	Utts	Hours	Trials
slr52_sinhala	si	478	185,293	224.5	12,000
slr127_tamil	ta	531 -> 638	89,401	150.1	12,000
kathbath_tamil	ta	60 -> 49	7,171	13.2	150,000 official
nisp_tamil	ta + en	65 (all bilingual)	4,905	13.3	8,000 + 4,000 cross-lingual
slr65_tamil	ta	50	4,291	7.1	12,000
slceleb2026_sinhala	si	123	46,960	85.1	real sessions

291,061 files converted to 16 kHz mono PCM_16, **zero conversion errors**. **Tamil speaker pool: 50 -> 706 as built** (802 after the QC corrections below). Note: the 17 August report quotes ``50 -> 752'', which reconciles with neither figure and should be corrected.

Three structural facts were established and matter for every claim downstream:

1. **Sinhala has almost no open audio**. SLR52 is the Sinhala open corpus. Sinhala is **absent from Common Voice** (verified against the CV 26.0 locale list, 294 locales, no si).
2. **Every large Tamil corpus is Indian Tamil, not Sri Lankan Tamil**. Excellent training data, useless for claiming Sri Lankan performance. The Indian-trained / Sri-Lankan-evaluated **dialect gap is an unpublished number this project is uniquely positioned to produce**.
3. **SiTa is CC BY-NC 4.0** --- non-commercial, so unusable in the VoiceID product path. The only such restriction in the collection and easy to forget.

6.7 The dataset quality audit --- two speaker-identity bugs [M]

A seven-layer QC pipeline (integrity, signal quality, label reliability, trial audit, power, calibration, fairness) was run over all corpora. It found two identity bugs **in corpora built and documented by this project**, both invisible to every check that does not use audio.

Bug 1 --- SLR127: the prefix is a numbering scheme, not a session

The original ingestion took field 2 as the speaker (531 distinct values, matching the documented speaker count) and the ISTL/MICI/MILE prefix as a recording session --- yielding ``107 multi-session speakers'' and the claim of the only genuine cross-session read-speech trials in the collection.

Embedding the centroids settled it:

Comparison	Cosine
same number, same prefix (split-half of one speaker)	0.961
same number, different prefix	0.269
different numbers (impostor baseline)	0.319

Cross-prefix pairs score **below the impostor baseline**. ISTL_0000202 and MILE_0000202 are two different people; each collection batch numbers its speakers from scratch.

	Before	After
Speakers	531	638
EER	55.31 % (worse than chance)	1.07 %
d' / NMI	3.14 / 0.963	5.23 / 0.984
silhouette	0.377	0.463

The old trial list drew 3,000 ``targets'' across prefixes --- pairs of different people labelled as the same speaker --- which is why it scored worse than chance.

Finding III-2. ``531 distinct values matches the documented 531 speakers'' was a red herring. Counting agreement is not identity verification; only audio can answer an identity question. Every label-quality metric moved in the direction a correct key predicts without being the quantity optimised --- about as close to independent confirmation as is available without ground truth.

Bug 2 --- Kathbath: recurring split ids are the same person

11 numeric ids occur in both valid and test_known and had been namespaced apart. Cross-split cosine: **0.992** (baseline 0.348). They are the same people. **49 speakers, not 60**. Training labels merged; official trial paths left untouched to preserve comparability.

Other findings from the audit

- **Channel is nearly identity in the read corpora.** $\eta^2(\text{SNR} \mid \text{speaker}) = \mathbf{0.91}$ in SLR127 and 0.68 in SLR52 --- a model can score recording conditions rather than voice. Correcting bug 1 raised this from 0.74 to 0.91, the expected direction (the confound was partly hidden by the bug).
- **The channel probe.** Multinomial logistic regression predicts **corpus-of-origin from the speaker embedding at 91.2 % accuracy** against 20 % chance. No single-corpus absolute EER should ever be quoted as a performance claim.
- **Trials do not buy power; speakers do.** Kathbath: 50,000 official trials, 20 speakers, **MDE 4.5--6.2 pp**. SLR127: 12,000 trials, 638 speakers, **MDE 0.26 pp** --- 17x better resolution from a quarter of the trials.
- **Kathbath's official lists contain 351 self-pairs** (a file paired with itself), which score cosine 1.0 by construction and bias the published EER low by ~ 1.4 % of targets. Left untouched deliberately --- altering them would break comparability with every published IndicSUPERB number --- but now quoted with the caveat attached automatically.
- **First cross-lingual number in the project.** NISP: same speakers score 1.10 % within Tamil and **2.90 %** Tamil $\leftrightarrow$ English --- a **2.6x** degradation from the language switch alone, on identical speakers and one embedding.
- **Fairness cannot be assessed on the two largest corpora** --- SLR52 and SLR127 ship no gender labels. Where measurable, $\text{FDR} \geq 0.97$ (no material gender disparity).

The same-recording confound, measured directly

SLCeleb 2026 V3 is the only corpus in the collection with genuine sessions (400 YouTube videos; 100 of 123 speakers span 2+):

Pair type	Mean cosine
Same speaker, same session	0.612
Same speaker, different session	0.474
Different speakers	0.136

Allowing same-session targets gives $d' = 3.46$; restricting to cross-session gives $d' = 2.44$. **The confound inflates apparent separability by 42 %**, on identical speakers and one embedding. Every other corpus maps one session per utterance, so their EERs carry optimism of about this size.

6.8 Benchmark v1 --- the open-set rebuild [M]

Three findings landed **before a single full experiment ran**, and all three changed how results must be produced.

6.8.1 Every shipped corpus list was 100 % closed-set

corpus	train spk	test spk	overlap
slr52_sinhala	478	478	478 (100 %)
slr127_tamil	638	638	638 (100 %)

slceleb2026_sinhala	123	123	123	(100 %)
slr65_tamil	50	49	49	(100 %)

Why this is worse than ``optimistic``. For a speaker seen in training, f has been explicitly optimised to collapse that speaker onto its AAM-Softmax centroid w_c , so the margin enforced at training time transfers directly into the trial score. **The inflation grows with capacity to memorise centroids** --- a 95 M SSL front end gains more from it than a 1.4 M ResNet. A closed-set ranking of architectures is therefore partly a ranking of memorisation capacity, and the two orderings are not the same.

Rebuilt: 70/10/20 by speaker, stratified by duration, deterministic, SHA-256 per list. Four corpora held out entirely as generalisation probes. **The shipped lists/ were left untouched** so prior results stay reproducible --- they are simply not comparable.

6.8.2 The power ceiling

With m trials per speaker, S speakers, intra-speaker correlation ρ :

$$n_{\text{eff}} = \frac{mS}{1 + (m-1)\rho} \rightarrow \frac{S}{\rho}$$

At $S = 91$, $\rho = 0.7$: $n_{\text{eff}} \leq 130$, giving **$\sim \pm 6$ pp on a single absolute EER**. Two consequences, both acted on:

1. **Trial lists cut 80k -> 20k pairs.** At $m \approx 20$ the design already reaches 98 % of the ceiling; generating more pairs is measurement theatre.
2. **Comparison must be paired.** Writing per-speaker error as $e_A(s) = \mu(s) + a(s)$, the speaker-difficulty term $\mu(s)$ is common to both systems on identical trials and **cancels in the difference**.

Validated against synthetic data with known ground truth: EER matched the repo's own tuneThreshold exactly; a known injected difference was correctly detected; a system against itself correctly found nothing; **paired intervals were 2.6x tighter** than absolute EER.

6.8.3 The research and production stacks were never the same model

voiceid (deployed) learns a **softmax weighting over all 13 WavLM layers**; the trainer reads only the **last** one --- the layer masked-prediction pretraining works hardest to strip speaker identity out of. **Nothing had measured the difference**. This became the study's principal result.

6.9 Stage F and the SSL layer probe --- the principal result [M]

Stage A's biggest surprise was that `ssl_wavlm` (94.98 M) finished **last** on both languages despite being 6x larger than the winner. A direct probe --- run the frozen encoder, mean-pool each hidden state, cosine-score, **no training at all** --- explains it:

Layer	si EER %	ta EER %
0 (CNN output)	22.30	11.05
3	29.27	15.78
6	36.50	24.48
9	38.90	31.68
12 (--ssl_layer -1, the default)	40.33	32.08

Monotonic degradation with depth in both languages. The conventional default is the **worst available choice** --- 1.81x worse (si) and 2.90x worse (ta) than layer 0.

Scientific explanation. Masked-prediction pretraining rewards recovering a masked frame from context, which drives upper layers toward phonetic content and treats speaker identity as nuisance. The last layer is the one the objective worked hardest to make speaker-invariant.

Front-end sweep, held-out test [M]

Front end	si	ta	Family
ssl_mhubert_lw	2.369	0.422	SSL, 13 learned layer weights
ssl_mhubert_ecapa	2.793	0.402	SSL + full ECAPA backbone
ssl_wavlm_ecapa	3.166	0.392	SSL + full ECAPA backbone
ssl_wavlm_lw	3.297	0.623	SSL, layer-weighted
ssl_wavlm_low (layer 0)	4.033	1.125	SSL, single layer
ecapa1024 --- mel baseline	4.375	0.884	filterbank
mfcc40d	4.396	1.135	cepstral
mfcc80	4.486	1.125	cepstral
ssl_wavlm_ft (unfrozen, single layer)	5.404	1.788	SSL, fine-tuned
ssl_wavlm_mid (layer 6)	5.525	1.898	SSL, single layer
ssl_wavlm --- layer 12	8.247	4.620	SSL, last layer

Finding III-3 (the programme's headline). How the SSL layers are read matters more than which architecture reads them. Same encoder, same data, same recipe: **8.247 -> 3.297 on Sinhala, a 2.5x improvement from 13 scalar parameters.** Against the strongest mel baseline, the layer-weighted SSL front end is **46 % better on Sinhala** and the SSL+ECAPA hybrid **55 % better on Tamil**. Paired bootstrap: si Delta = -2.006 pp, CI [-2.420, -1.550], p = 0.000.

Learned weighting beats every fixed depth (lw > layer 0 > layer 6 > layer 12), so the gain is not merely ``use a low layer" --- **the mixture carries information no individual layer does.**

Multilingual pretraining pays at equal capacity. mHuBERT-147 (94.97 M, 147 languages including si and ta) beats WavLM-base-plus (94.98 M, English) on both languages: si Delta = -0.927 pp (CI [-1.294, -0.654], p = 0.000), ta Delta = -0.191 pp (p = 0.005). Because the parameter counts are matched to within 0.01 %, this **isolates pretraining coverage from capacity** --- a clean contrast that the confounded XLS-R-300m comparison could not provide.

A pre-registered prediction, falsified and kept

The 12 August design predicted speaker information would peak **mid-stack**, and ssl_wavlm_mid was built at layer 6 on that basis. Layer 6 measures 36.5 % --- barely better than the last layer, far off layer 0's 22.3 %. **The prediction was wrong.** The entry was retained as the mid-curve data point and the falsification recorded in the registry **rather than edited away**. This is what pre-registration is for.

A second pre-registered prediction, also falsified

MFCC was predicted to lose badly: the DCT decorrelates filterbank channels, which is essential for diagonal-covariance GMM-UBM systems but actively harmful for a CNN/TDNN that wants the inter-band correlations where formant structure lives. **Measured: MFCC lands within ~0.2 pp of log-mel** and the paired contrasts are not significant on Sinhala (mfcc80 vs ecapa512: Delta = -0.202, p = 0.463). The prediction was directionally right and practically wrong.

6.9.1 A new result: the last-layer penalty scales with linguistic distance

Language	SSL last layer	mel ECAPA-1024	penalty
English (en_matched)	7.880	7.580	1.04x
Sinhala	8.247	4.375	1.88x
Tamil	4.620	0.884	5.23x

Finding III-4. Last-layer SSL practice is an English-centric default whose cost is a function of the target language. WavLM is pretrained on English, so its upper layers specialise toward English phonetic content; discarding the lower layers removes speaker information that the English task can partly recover from context and the Sri Lankan languages cannot. **Without the English control this would have looked like an SSL failure rather than a transfer failure** --- the direct justification for running the control arm.

6.10 Stage A --- architecture sweep, and the capacity null [M]

Seven backbones x {si, ta}, everything else pinned (embedding dim 256, 80 mels, 2.0 s train / 3.0 s eval crops, identical MUSAN+RIR augmentation, Adam 1e-3, 60 epochs, AS-Norm off during training).

Architecture	si test	ta test	Params
ecapa512	4.285	1.045	5.99 M
ecapa1024	4.375	0.884	14.46 M
resnetse34v2	4.708	1.376	7.37 M
mlpmixer	4.930	1.256	7.71 M
resnetse34l	4.990	1.708	1.40 M
vggvox	5.847	1.949	3.64 M
ssl_wavlm (layer 12)	8.247	4.620	94.98 M

Finding III-5. This regime is data-limited, not capacity-limited. ECAPA-512 matches ECAPA-1024 at 40 % of the parameters: si Delta = 0.091 pp, CI [-0.352, +0.415], **p = 0.835**; ta Delta = -0.151 pp, p = 0.134. **2.4x the parameters buys nothing measurable.** This is a null result with a tight interval, not an absence of evidence, and it directly justified not pursuing larger from-scratch models.

MLP-Mixer, revisited. Phase I's architecture was carried into the v1 sweep on equal terms and placed **4th of 7 on Sinhala and 4th of 7 on Tamil** --- competent but beaten by ECAPA at both sizes, and beaten decisively by every layer-weighted SSL front end. It is a fair, open-set, held-out verdict on the Phase I architecture: **respectable, not competitive.**

6.11 Stage H --- does the backbone still matter under SSL features? [M]

Stage A varies the backbone with the front end fixed; Stage F varies the front end with the head fixed. Stage H measures the interaction neither can see.

System	si	ta
ssl_mhubert_lw (pooling head)	2.369	0.422
ssl_mhubert_ecapa (full ECAPA backbone)	2.793	0.402

The answer is language-dependent, which is itself the finding. On Sinhala the heavier backbone hurts significantly (Delta = -0.423 pp favouring the pooling head, p = 0.001); on Tamil

the difference is **not significant** (Delta = 0.030, p = 0.406). At 0.392 % (ssl_wavlm_ecapa), Tamil is the best single result in the programme.

Practical reading: **once features come from a well-read SSL encoder, representation dominates architecture.** Spend effort on the front end.

Caveat recorded in the module: ECAPA's dilations (2, 3, 4) were tuned for 100 Hz mel frames; SSL encoders emit **50 Hz**, so the receptive field spans twice the intended time. Not necessarily harmful, but it travels with the front-end change and must not be attributed to it.

6.12 Domain, channel and corpus effects [M]

Read speech vs genuine cross-session audio

si_pooled trains on both Sinhala domains and scores each separately, so one model gives a like-for-like comparison:

Backbone	read (slr52)	wild (cross-session)	ratio
ecapa1024	3.297	16.501	5.0x
ecapa512	3.357	16.978	5.1x
resnetse34v2	3.811	18.887	5.0x
mlpmixer	4.275	17.638	4.1x
ssl_wavlm	7.965	27.508	3.5x

Finding III-6 --- the most important caveat in the programme. Genuine cross-session audio is 4--5x harder than read speech for every backbone. Any deployment claim resting on read-speech numbers is overstated by roughly a factor of five --- **including every v0 number, which used SLR52 throughout.** The architecture ranking is stable across domains, which is reassuring for external validity even where absolute numbers do not transfer.

The Tamil channel artefact

SLR127 pools three collection sites. Splitting impostor trials on whether both sides come from the same site:

Impostor type	Count	Mean score	EER
cross-batch	4,805	-0.0433	0.475 %
same-batch	5,151	+0.0527	1.298 %
all (as reported)	9,956	---	0.894 %

Cross-batch impostors score negative --- rejected almost for free because the channel differs, not the speaker. **~48 % of the trial list is that easy kind.** Reproduced across architectures at 1.26--3.20x inflation.

Reporting rule adopted: quote the same-batch figure for SLR127. Tamil remains substantially easier than Sinhala after control (1.30 % vs 4.38 % for the same backbone) --- **the asymmetry is real; its magnitude was inflated.**

English control --- the ranking does not transfer

Backbone	English en_matched	Sinhala	Tamil
resnetse34v2	7.260	4.708	1.376
ecapa1024	7.580	4.375	0.884
ssl_wavlm (last)	7.880	8.247	4.620
vggvox	10.420	5.847	1.949

On English, resnetse34v2 edges ecapa1024 and last-layer SSL is competitive. On Sinhala and Tamil, ecapa1024 leads and last-layer SSL collapses. **An architecture ranking established on English does not transfer to these languages** --- the direct answer to the question that motivated the English arm.

The en_full positive control reached **3.496 % validation EER** on VoxCeleb1-O, close enough to the known reference for this recipe to validate the harness end to end --- loader, augmentation, scoring and checkpoint selection.

A documented negative result

Trained alone on 82 speakers, si_celeb reached **19.7--27.3 % EER** across every architecture with minDCF ~ 0.99 . Training worked (loss 7.34 $\rightarrow$ 0.66, train accuracy 6 % $\rightarrow$ 85 %): **the models learned their training speakers and transferred to none**. Two design faults --- 82 speakers is far too few, and the validation set has 9 speakers ($n_{\text{eff}} \leq 13$), which cannot measure anything yet selected every checkpoint. Retained as a negative result; its EERs should not be quoted. Its correct use is as an evaluation probe, where it is highly informative.

6.13 The evaluation-integrity incident --- 22 August [M]

Before the v1 benchmark could be closed, four defects were found between the 17 and 22 August states. **Three of them would have produced confident, plausible, wrong numbers rather than an error.**

#	Defect	Consequence had it stood
1	Validation dataloader deadlock in two	Stuck since 16 Aug, 6 attempts; Stage H would have been reported as 2/4
2	ssl_wavlm_ecapa runs si_pooled keyed on condition name, not on data	All seven pooled-Sinhala runs unevaluable; the read-vs-wild contrast lost entirely
3	--ssl_* arguments absent from resolved_parameters	Evaluation rebuilt SSL models from model defaults : ssl_layer fell back to -1. F_ssl_wavlm_low, trained on layer 0, would have been scored on layer 12 --- every tensor matching, no warning
4	Trainer argparse defaults not applied at rebuild	--encoder_type defaults to SAP in the trainer, ASP in the model class. Nine evaluations reported EERs from partly-random weights

The root cause, named precisely: the trainer passes its entire parsed namespace to the model, so a flag nobody mentions still reaches the model carrying the trainer's default. Any evaluator rebuilding from a manifest inherits the model's defaults instead, and **the divergence is silent whenever it does not change a tensor shape.**

Fixes: arguments recovered from `command.json` (the literal `argv`); `trainSpeakerNet.py` AST-parsed for **all 82 `argparse` defaults**, closing the whole class rather than the two flags observed; **a skipped backbone tensor is now a hard failure, not a warning** (``a wrong number is worse than no number''); prior evaluations archived before overwrite. **All 54 experiments now rebuild with zero backbone mismatch.**

7. The three parallel strands (August 2026)

7.1 Proposals A1--A9 --- literature designs, matched comparison [M]

Nine architectures were drawn from a 32-paper literature study and implemented as self-contained code that **never modifies voxceleb_trainer/** (asserted by selftest.py via git status --porcelain; 13 checks, 0 failures). Each run copies the **literal argv** of its matched v1 baseline and overrides only what the proposal changes, so the proposal is the single changed factor.

A constraint discovered, not designed

DomainTable reported that the si and ta conditions each contain **one language and one corpus**. A1, A6, A7 and A8 all depend on a domain variable --- adversarial, disentangling, or per-language. With one class, every discriminator is constant, every gradient-reversal term is uninformative, per-language vectors collapse to one, **and each method silently degrades to its own baseline while still producing a number**.

Finding III-7. Four of nine proposals are undefined on the conditions the benchmark was built from, and nothing in their papers or READMEs says so.

A7 is the cautionary case, because the guard was added only after it ran. Trained on ta with n_languages left at its default of 3, it built three weight vectors and a three-way language-ID head on data containing one language, trained to completion, and scored **0.663 % against its baseline's 0.623 %**. That reads as a small regression by a plausible margin. **It is not a regression** --- it is shared layer weighting plus two dead vectors and a degenerate head, measuring nothing. The run was retired with that explanation rather than deleted.

A method that is undefined on the data still trains, still converges, and still emits a number in the right range. Only a guard derived from the data catches it.

Results

Proposal	Design and rationale	Predicted [P]	Measured [M]	Verdict
A3 SL-AMEC	Training-free calibrator conditioned on trial metadata (language, corpus, set)	metadata conditioning improves EER	Cllr 0.8356 -> 0.1873 (median 4.5x, 1,225 params); 0 / 33 systems significant on EER; 29 / 33 show < 1e-9 EER change under shuffled metadata	Split: large positive on calibration, clean null on metadata

Proposal	Design and rationale	Predicted [P]	Measured [M]	Verdict
A6 channel-aware SSPS	Project the channel subspace out before clustering, on the premise that SSPS harvests same-channel rather than same-speaker pairs	correction improves cluster purity	plain SSPS speaker purity 0.9302 / NMI 0.9810 vs channel-aware 0.9099 / 0.9767	Premise fails here; correction slightly harmful
A9 ARI-SubCenter	Sub-centre AAM with K set per corpus from a measured label-reliability statistic rather than fixed at K=3	sub-centres absorb label noise the QC audit proved present	si 4.375 -> 3.962 , Delta = -0.413 pp , CI [-0.677, -0.015], p = 0.038 ; ta 0.884 -> 0.934, p = 0.668	First positive result --- and mechanistically coherent
A1, A2, A4, A5, A7, A8	---	---	training / queued / retired	No EER claimed

Why A3's null is strong rather than merely underpowered. A null from the bootstrap alone would be ambiguous at 18--26 test speakers. A **bit-identical result under metadata shuffling** is not ambiguous --- the metadata is inert. The metadata-only negative control sits at 98--99 % EER, confirming the harness measures what it claims.

Why A9's positive is more than a number. The asymmetry follows the QC audit exactly. SLR127 (Tamil) is the corpus whose identity error the audit **found and corrected** (638 speakers had been collapsed into 531). SLR52 (Sinhala) is the corpus the audit **flagged as unresolved** --- its profile is consistent with account-level rather than person-level identity, and no metric can settle that without a listening pass. Sub-centres are the standard response to residual within-class identity heterogeneity, and they improved the corpus where such heterogeneity is suspected but uncorrected while doing nothing measurable where it was found and fixed. Held as **promising, not settled**: p = 0.038 with an interval reaching -0.015 pp, single seed.

What the proposals strand has already redirected

- **Calibrate, do not annotate.** 4.5x better Cllr for 1,225 parameters; the metadata plumbing A3's full form requires is not worth building.
- **The channel confound is real but shallow.** v1 showed channel is linearly decodable (91.2 % probe accuracy); A6 shows it does **not** dominate local neighbourhood structure. Work aimed at the confound should target the **scoring** stage, not the clustering stage.
- **The clearest data priority in the project.** Four of nine methods need a domain variable the collection supplies only by pooling corpora, which confounds domain with language. **Acquiring within-speaker, cross-channel Sri Lankan recordings would unlock four of nine proposals at once** --- a sharper priority than any individual method.

7.2 transformer_sv T1--T6 --- from-scratch transformers [M]

The framing, which decides what is worth running. v1 already uses transformers --- WavLM and mHuBERT are transformer encoders, and reading them through learned layer weights gives the benchmark's best systems. What v1 never trained is a transformer **speaker encoder**: every from-scratch backbone in the benchmark is convolutional.

Pre-registered prediction [P], recorded before any run: from-scratch transformers will lose to ECAPA-TDNN on these corpora. Justification: v1 Finding III-5 showed the regime is data-limited, not capacity-limited ($p = 0.835$ for a 2.4x parameter increase), and transformers are the more data-hungry family. **A clean negative would be the measured argument for the SSL route on low-resource languages, on the same trials as everything else.**

T5 --- the cost result (training-free, CPU) [M]

All four attention arms hold **11,062,848 parameters exactly** (verified), so nothing below is a capacity effect.

input	tokens	interaction MACs	projection MACs	interaction share
2 s (training crop)	50	7,680,000	78,643,200	8.9 %
3 s (evaluation)	75	17,280,000	117,964,800	12.8 %
8 s	200	122,880,000	314,572,800	28.1 %

The fitted log-log MAC exponent over 1--8 s is **1.134, not 2**.

arm	interaction MACs @ 2 s	vs full	measured latency @ 2 s
full	7,680,000	---	15.7 ms
window (+/-8)	2,611,200	0.34x	16.4 ms (+4 %)
pooled (stride 4)	1,996,800	0.26x	16.5 ms (+5 %)
linear	9,830,400	1.28x	17.1 ms (+9 %)

Finding III-8. The $O(T^2)$ objection does not apply at these input lengths. With x4 convolutional subsampling a 2 s crop is 50 tokens, so the quadratic term is a minority of attention's own cost, which is itself a minority of the model's. **linear attention is 1.28x more expensive than full attention here** --- its crossover is at $T \sim 2 * d_{\text{head}} = 128$ tokens ~ 5.1 s --- and **every ``efficient'' arm measured slower in wall-clock**. Consequence: the accuracy arms use **full** attention, which is also the only mode that is a control (window with $w \geq T-1$ is bit-identical to it, asserted).

T1 --- the first from-scratch transformer accuracy result [M, NEW]

This result completed on 24 August and has not appeared in any prior report.

MFAConformer (Conformer encoder + multi-scale aggregation, full attention), parameter-matched to its baseline:

System	Params	si test EER (cosine)	si test EER (AS-Norm)	minDCF
A_ecapa512 (baseline)	5,994,688	4.285	4.224	0.303
MFAConformer_full_si_s42	6,135,504	4.950	4.718	0.326
Difference	+2.3 %	+0.665 pp	+0.494 pp	+0.023

The pre-registered prediction is confirmed. A from-scratch Conformer, at matched parameters and under an identical recipe on identical trials, is **~16 % relatively worse** than a convolutional ECAPA-TDNN on 336 Sinhala training speakers. Combined with Finding III-3 --- where a pretrained transformer read through learned layer weights reaches 2.369 % on the same trials --- this gives the programme a clean, self-contained argument:

On low-resource languages the value of a transformer is in its pretraining, not in its architecture. The same model family loses by 0.67 pp when trained from scratch and wins by 1.92 pp when pretrained and correctly read.

Caveat: single seed, one language. The paired bootstrap has not been run on this contrast, and at $S = 91$ the +0.665 pp gap is near the resolution limit established in section 6.8.2. Treat as directional pending the ta arm and a seed replication.

Two numerical defects found by these launches [M]

1. **fp16 attention overflow.** $q @ k^T$ is the one product in a transformer with no bound on its magnitude. In fp16 it overflows to inf, and $\text{softmax}(\text{inf} - \text{inf})$ is NaN. Under --mixedprec this killed MFAConformer on Tamil at ~9k steps **with no error message** --- the loss simply printed nan. ECAPA-TDNN has no such product, which is why the same recipe never exposed it. Fixed by computing scores, position bias and softmax in fp32 even under autocast.
2. **fp16 mel front-end overflow --- worse, and unrelated to attention.** Measured over 30,000 real augmented samples: MUSAN mixing and RIR convolution push the waveform to $|x| = 5.66$, and the fp16 power spectrogram peaks at **30,592 against fp16's 65,504 ceiling --- 2.1x headroom, not orders of magnitude.** One inf there poisons **Batch-Norm's running statistics permanently**, because those are written in the forward pass and GradScaler only guards the backward. Eval mode then returns NaN for every input **while training accuracy still looks healthy** (TAcc 23.59 % and TLOSS nan in the same epoch), and the run dies at its next validation.

The diagnosis generalised beyond its own strand. The v1 benchmark is clean (**0 of 54 runs** have a NaN training-loss epoch), but **two A-series proposal runs** --- A5_dk_campp/DKCAMPP_ta and A4_ls_cam/LSCAM_ta --- fail with the identical traceback. Both are CAM++-family models that build their **own** MelSpectrogram inside the model rather than taking the trainer's. All eight existing A-series checkpoints were then verified clean (0 poisoned BN buffers, 0 NaN weights), so **no published A-series number is affected** --- the two runs simply never produced one, and the one-line fix they need is recorded.

An initial reading was wrong and was corrected within the hour. The first diagnosis held the NaN losses to be a cosmetic averaging artefact ("a blemish in the record, not a defect in the model"). Both affected arms then died at their next validation. The superseded reasoning was **kept in the document and marked superseded** rather than deleted --- the record shows the wrong hypothesis, the evidence that killed it, and the correct one.

Cost of the detour: ~25 GPU-minutes of aborted training and ~15 of probes, against the ~80 GPU-hours the first dispatch would have spent producing one NaN arm and four arms trained under numerically broken attention.

A recipe deviation was considered and rejected on evidence. The obvious reading --- "the ECAPA recipe cannot train a transformer, it needs warm-up" --- would have made every T-series number incomparable with the baselines. Probes showed lowering the learning rate to $1e-4$ is **measurably worse** on both languages, so the arms stayed protocol-faithful with no deviation.

Status: T1 si complete; T1 ta, T1 no-attention, T2 tf4/tf0 training; T3, T4, T6 queued. scaffold.py --check reports what has produced numbers and what has not, and is the figure to trust.

7.3 Feature-level testing --- hand-crafted descriptors under a byte budget [M]

A separate report had proposed four hand-crafted acoustic frameworks for ultra-low-payload verification (a QR-code-carried speaker template). **None had ever been tested in this project.** All four were implemented and run on real Sinhala and Tamil data, matched by construction: same speaker count (40 each), same trial count, same 4 s crop, identical back end, and **100 % of genuine pairs cross-session.**

System	bytes	Sinhala EER %	Tamil EER %	si minDCF
formant trajectories (the report's top pick)	40	43.07	39.50	0.997
formant (static, control)	8	35.90	31.43	0.994
glottal (IAIF + LF)	23	19.73	13.80	0.989
LPC-12 LSP	24	18.00	12.13	0.934
bispectrum (HOSA)	40	30.67	29.70	0.999
all four fused	127	17.30	12.93	0.870
hybrid micro-vector (the report's recommendation)	191	10.07	4.47	0.737
ECAPA-TDNN (PCA-128)	128	3.63	0.57	0.273

Findings, each with its control:

1. **The report's ranking inverts under measurement.** Formant trajectories, rated ``High / Extremely High'', are the **worst** of the four and barely separate speakers (minDCF ~1.00 means the detector is worthless at a 1 % prior). LSP, rated lowest, is **best --- in the smallest payload of any system tested.**
2. **The cubic-trajectory descriptor is worse than averaging.** A static control derived analytically from the same coefficients uses **8 bytes instead of 40** and scores **7.2 pp (si) / 8.1 pp (ta) better.** Formant extraction is not the problem --- the selftest recovers a synthetic four-resonance vocal tract to within 1 %.
3. **The byte budget was never the bottleneck.** 8-bit quantisation costs ≤ 0.07 pp anywhere, and several systems improve slightly (mild regularisation). **The report's core premise holds; its feature choices do not.**
4. **The recommended hybrid is a net loss at every mixing weight.** Sweeping the physical-feature weight from 0 to 0.7 is monotone in the tail; at the specified equal weighting it costs **+6.4 pp (si) and +3.9 pp (ta).** The control that makes this attributable: **PCA-truncating ECAPA from 192 B to 128 B costs nothing** (3.63 / 0.57 vs 3.63 / 0.60), so the loss is entirely the fusion, not the compression.
5. **The phonetic-degradation premise is weak.** Tamil carries 2.0x the retroflex and 3.4x the geminate load of Sinhala, but Sinhala's prenasalized stops --- the category the report leans on hardest --- occur **0.47 times per 100 characters**, roughly once every 200 characters. Splitting trials by phonetic density: Tamil goes the predicted way on 5 of 5 systems, **Sinhala goes the wrong way on 3 of 5**, and every effect is small relative to the bootstrap intervals.
6. **The QR mapping half-checks out.** 192 bytes is exactly version 10 (57x57) at EC=M --- the claim holds. But the report's own **512-byte ceiling needs version 18 (89x89)**, outside its stated range, and at a 2 cm print size its modules fall to **0.206 mm**, below the ~0.25 mm practical floor for a phone scan. **A 512-byte payload and a 2 cm code are not simultaneously achievable.** Dropping to 128 B also drops the symbol to version 8 and raises module pitch to 0.351 mm --- **14 % more optical margin, for free.**

A confound identified and reported rather than buried. Tamil is easier on every system, but **the gap grows with how good the system is** (1.03x on bispectrum, 6.41x on ECAPA). A language-intrinsic advantage would shift all systems together; what behaves this way is head-room. Measured audio quality confirms Tamil is cleaner on all four periodicity measures (HNR 3.77 vs 3.00 dB, jitter 0.036 vs 0.057). **``Tamil verifies 6x better than Sinhala'' must not be quoted as a language finding.**

And it independently reproduced Finding III-1. A single shared threshold at the pooled EER point costs Tamil **+1.65 pp** --- nearly 3x its own EER --- and pushes Sinhala's miss rate to 6.1 %. Per-language calibration is mandatory, now demonstrated on hand-crafted and deep features alike.

8. Failure register --- every failed result and what it produced

This section exists because in this programme the failures carry more information than the successes, and several of them are the publishable contributions.

#	What failed	Predicted	Actual	Root cause established	What was identified
F1	MLP-Mixer V1 (MSE distillation)	10--11 %	16.13 % --- worse than no distillation	MSE on L2-normalised embeddings is ~ 0.0002 ; distillation was 0.008 % of the loss	Finding I-1: embedding distillation requires an angular loss. Geometric, not empirical --- transfers to any normalised-embedding distillation
F2	MLP-Mixer V2_Large (alpha 0.7)	11--12 %	14.84 % --- worse than a 3x smaller model	Student (7.84 M) exceeded teacher (3.87 M) by 102 %; alpha=0.7 pinned it to a lower-capacity teacher	Finding I-2: optimal alpha tracks capacity ratio. Better teacher alignment can coexist with worse verification --- alignment is not the objective
F3	NestedSpeakerNet 3 attempts	Net 1.3 % EER, 2x faster	NaN x2, or 87 % worse	Anti-correlated inter-level features ($\rho = -0.23$) make nested aggregation amplify rather than regularise; Hessian condition $\sim 160,000$	Finding I-3: five measurable domain-compatibility criteria; audio meets 0 of 5. Predicts failure for video action recognition, ASR, time-series too

#	What failed	Predicted	Actual	Root cause established	What was identified
F4	The whole Phase I strategic direction	competitive lightweight SV	~12x off SOTA at matched size	Training small models from scratch on VoxCeleb is a solved-and-lost race; the field fine-tunes released checkpoints	Redirected the entire programme (audit R1). ReDimNet-B1: 0.85 % at 2.2 M vs 10.32 % at 2.66 M
F5	English P0 baseline	a reference EER	Never ran. 8 launch attempts 2026-05-27, empty scores, no checkpoints	564 missing VoxCeleb2 wavs	Fixable in one filter operation (audit C1); the blocker was catalogued, not diagnosed, for ~5 weeks
F6	Jan--Jun 2026 experimental claims	7 architectures + SSL + deployment	No supporting artefact for any of it	Not established	The programme's evidence standard was rebuilt around this. See section 5
F7	PEFT-beats-full-FT (UniPET-SPK)	LoRA >= full FT	full FT beat LoRA at every seed, 2.2x	Closed-set eval removes full FT's overfitting-to-source penalty	Falsification reported with mechanism and with the arbiter experiment named. Still unrun --- see section 10
F8	Mid-stack SSL layer peak	speaker info peaks ~layer 6	Monotonic decay with depth; layer 6 barely beats layer 12	Masked-prediction pretraining makes upper layers speaker-invariant by design	Finding III-3 , the programme's headline. The falsified entry was kept as the mid-curve data point

#	What failed	Predicted	Actual	Root cause established	What was identified
F9	MFCC will lose badly to log-mel	large loss from DCT decorrelation	within ~0.2 pp , not significant on si (p = 0.463)	Prediction was directionally right, practically wrong	Pre-registration works: an intuition that felt certain was cheap to test and false
F10	PLDA under domain shift (v0)	PLDA > cosine	Hurt Sinhala for every model	PLDA fit dominated by 478 near-single-session si speakers -> within-speaker covariance underestimates channel variability	Flagged as a benchmark artefact hypothesis, not a refutation --- and later vindicated: after fine-tuning moved the system in-domain, PLDA became redundant exactly as the literature predicts
F11	si_celeb trained alone (82 spk)	a usable in-the-wild system	19.7--27.3 % EER	82 training speakers far too few; 9-speaker validation set selected every checkpoint but can measure nothing	Retained as a documented negative result. Correct use is as an evaluation probe, where it is highly informative

#	What failed	Predicted	Actual	Root cause established	What was identified
F12	A6 channel-aware SSPS	projection improves clusters	speaker purity fell 2.0 pp	Clusters were already speaker-shaped (speaker NMI 0.98 vs corpus NMI 0.28)	Sharpened v1's channel finding: channel is linearly decodable but does not dominate local neighbourhoods. Bounded honestly --- the bootstrap was supervised, biasing toward this result
F13	A3 metadata conditioning	metadata improves EER	0/33 significant; 29/33 bit-identical under shuffling	The gain is entirely score-domain	A publishable controlled null : the shuffled-metadata control makes it inertness, not absent power. Stops the next group repeating it
F14	A7 PLLW first runs	per-language layer weighting	0.663 % vs 0.623 % baseline --- reads as a plausible small regression	Ran on single-language data with n_languages=3: two dead vectors and a degenerate head	Finding III-7 : methods undefined on the data still train, converge, and emit plausible numbers. Guards must derive from the data

#	What failed	Predicted	Actual	Root cause established	What was identified
F15	Efficient attention (T5)	cheaper than full attention	linear is 1.28x more expensive; all arms slower in wall-clock	x4 subsampling makes a 2 s crop 50 tokens; crossover is at 128 tokens ~ 5.1 s	Killed four planned training runs before they cost GPU-hours. A cost result that saved compute
F16	From-scratch Conformer (T1)	predicted to lose	lost by 0.665 pp to matched ECAPA-512	Data-limited regime (Finding III-5); transformers are more data-hungry	Prediction confirmed. Completes the argument: transformer value is in pretraining, not architecture
F17	Formant-trajectory descriptor	best of four frameworks	43.07 % / 39.50 % --- near chance	Averaging polynomial coefficients over phonetically uncontrolled segments destroys more than it captures	An 8-byte static control beat the 40-byte trajectory by 7--8 pp. The proposal is worse than the trivial alternative
F18	Hybrid micro-vector	deep + physical > deep	+6.4 pp (si) / +3.9 pp (ta) at the specified weight	Fusion, not compression --- PCA to 128 B costs nothing	Spend the whole budget on the deep embedding; 128 B also yields a more scannable QR symbol

8.1 Engineering defect classes --- the meta-finding

Across the year, **roughly 40 defects** were found and fixed. Their distribution is the point:

Class	Examples	Why it matters
Silent wrong numbers (the dominant class)	Checkpoints matching 0 tensors and reporting a believable ~41 % EER; int8 label overflow corrupting every EER past 127 targets; SSL models scored on the wrong layer; nine evaluations from partly-random weights; A7's degenerate run	None of these crash. Every one produces a plausible number. This is the failure mode the programme has paid for most
Numerical	fp16 attention overflow; fp16 mel front-end overflow poisoning BatchNorm permanently	Kill runs with no error message; the second also explains two unrelated A-series failures
Experimental hygiene	Smoke checkpoints silently resumed by full runs; English conditions leaking into every evaluation (+2.5x cost); Stage F front ends leaking into Stage A; manifest clobbering across invocations	Corrupt the comparison rather than the code
Infrastructure	4 cluster-wide reboots in 5 days; queue consuming rc=255 jobs without retry; pkill -f over ssh killing its own shell; slots blocked 5.5 h by lingering pty clients	Cost time, not correctness --- no training work was lost , thanks to per-epoch checkpointing and recorder-resume
Throughput	7 concurrent jobs collapsed every job to 5-16 Hz (114 dataloader processes saturating one NFS server, GPU ~0 %, CPU 86-93 % idle)	Capping concurrency at 3 restored 224--1,625 Hz, a 30--100x recovery . Fewer concurrent jobs finish the work strictly faster here

Finding III-9. In this programme the modal defect does not crash --- it returns a plausible number. Every methodological device adopted in Phase III (hard-failing evaluators, argv copying rather than manifest reconstruction, exact-reduction controls, shuffled-metadata negative controls, isolation assertions in selftest.py, data-derived guards) exists to convert a silent wrong number into a loud failure.

9. Prediction scoreboard

Pre-registered or explicitly stated predictions, scored against measurement.

#	Prediction	Source	Outcome	Verdict
1	LSTM+AE improves 20--35 % over baseline	Phase I	37 % (15.48 -> 9.68)	Met
2	MLP-Mixer student reaches 10--11 %	Phase I	16.13 % (V1), then 10.32 % (V2)	Failed, then met after diagnosis Failed
3	Larger student closes the teacher gap	Phase I	14.84 %, then 10.11 % --- never beat the small student	
4	Nested learning: -8--13 % EER, 2x faster	Phase I	NaN x2; 1.09x faster at best	
5	Cross-lingual degradation is 2--4x	CN-Celeb precedent	ta 2.3--4.0x; si 4.9--8.0x	Met for ta, exceeded for si Met
6	AS-Norm gives 20--30 % relative	Matejka 2017	12--39 % (5/6 cells in band)	
7	PLDA beats cosine under shift	arXiv:2204.11403	Hurt si; helped ta 2/3	Refuted at v0, vindicated in-domain Met, larger than expected
8	Cross-lingual score shift is systematic	arXiv:2110.09150	Cllr inflation up to 5.7x	
9	PEFT >= full FT under language shift	UniPET-SPK	full FT won at every seed, 2.2x	Falsified (closed-set caveat named) Partly met
10	Frozen encoder + head captures most SSL gain	WavLM paper	frozen 3.74 vs full 1.63 --- within ~1 EER of best zero-shot	
11	SSL speaker info peaks mid-stack	this project, 12 Aug	Monotonic decay with depth	Falsified --- and became the headline finding Falsified
12	MFCC will lose badly to log-mel	this project	Within ~0.2 pp, not significant	
13	Multilingual pretraining pays at equal capacity	this project	mHuBERT beats WavLM on both (p = 0.000 / 0.005)	Met
14	Regime is data-not capacity-limited	this project	ecapa512 = ecapa1024, p = 0.835	Met
15	Efficient attention will be cheaper	general practice	linear 1.28x more expensive at 2 s	Falsified

#	Prediction	Source	Outcome	Verdict
16	From-scratch transformers lose to ECAPA	this project, pre-registered	4.950 vs 4.285 (si)	Confirmed
17	Sub-centres help where label noise is uncorrected	this project (A9)	si -0.413 pp p=0.038; ta null	Met, mechanistically
18	Metadata conditioning improves calibration+EER	A3 source	Calibration yes (4.5x), metadata inert	Split
19	Channel dominates SSPS neighbourhoods	A6 source	speaker NMI 0.98 vs corpus 0.28	Falsified in this setup
20	Hand-crafted descriptors are viable sub-kB	feature report	best is 5x/21x worse than ECAPA at half the size	Falsified

Score: 8 met, 7 falsified, 3 split/partial, 2 failed outright.

A ~40 % falsification rate on pre-registered predictions is a healthy sign, not a poor one: it means the predictions were specific enough to be wrong, and that they were scored rather than rationalised. Two of the falsifications (11 and 16) are the strongest results in the programme.

10. What is safe to claim as of 31 August 2026

10.1 Safe

Every number in sections 6.9--6.12 is **held-out test EER on speaker-disjoint splits**, from a model verified to match its own checkpoint, with model selection made on validation trials only and the test set scored once.

Claim	Evidence
Best Sinhala system: ssl_mhubert_lw, 2.369 % cosine / 2.178 % AS-Norm , minDCF 0.156	v1 held-out test
Best Tamil system: ssl_mhubert_ecapa, 0.402 % cosine / 0.321 % AS-Norm	v1 held-out test
Layer-weighted SSL beats the mel baseline by 46 % (si) / 55 % (ta)	paired bootstrap, $p = 0.000$
Speaker information in WavLM decays monotonically with depth ; the conventional default is the worst choice	layer probe, no training involved
The last-layer penalty scales with linguistic distance (1.04x en, 1.88x si, 5.23x ta)	v1 + English control
Multilingual pretraining pays at matched capacity	mHuBERT vs WavLM, $p = 0.000 / 0.005$
The regime is data-limited, not capacity-limited	ecapa512 vs ecapa1024, $p = 0.835$
Cross-session audio is 4--5x harder than read speech	si_pooled, one model, both domains
~48 % of a published Tamil impostor list is rejected on channel, not identity	batch-split diagnostic, 3 architectures
Per-language calibration is mandatory	Cllr inflation 5.7x; shared threshold costs ta +1.65 pp
AS-Norm is the best cost/benefit lever in the project	12--39 % relative, training-free
A9 sub-centres help Sinhala (-0.413 pp, $p = 0.038$)	paired bootstrap, single seed
A3 metadata conditioning is inert over score calibration	29/33 bit-identical under shuffling
Two speaker-identity bugs found and corrected in corpora this project built	audio-based identity test

10.2 Not yet safe

- **Single-seed results throughout v1, the proposals, and the T-series.** The power ceiling binds: a single absolute EER carries $\sim \pm 5\text{--}6$ pp at $S = 91$. Differences of 0.09 pp (ecapa512 vs ecapa1024) are **not** resolvable; differences of 2.0 pp (SSL vs mel) **are**.
- **Any absolute EER quoted from a single corpus.** With $\eta^2(\text{SNR} \mid \text{speaker})$ at 0.68--0.91 and a 91.2 % channel probe, part of what is measured is the recording setup.
- **Any Sri Lankan Tamil claim.** Every large Tamil corpus in use is Indian Tamil.
- **T1's transformer contrast**, pending the Tamil arm and a paired bootstrap.

10.3 Must be corrected or withdrawn

1. **All January--June 2026 experimental results** (section 5). They cannot enter the thesis, any paper, or any future progress report.

2. **The nested-learning ``9.84 % EER, validated for efficiency" claim** in particular --- the measured record is three NaN collapses.
3. **The Phase I 10.32 % headline** --- unreplicated, single seed, predates --deterministic, and carries an unresolved internal conflict (14.62 % in one log) plus a suspected duplicated result set. Audit item R7 applies: re-run 3 seeds or retire the claim.
4. **The Period 2 report's ResNetSE34L specification** (34 layers / 6.8 M / ~8--10 % EER) --- the logs give 1.50 M and 15.48 %.
5. **The ``Tamil pool 50 -> 752" figure** in the 17 August report --- the measured build is 706 (802 after QC corrections).
6. **Every v0 EER** should carry the ~5x read-vs-wild optimism factor retrospectively, and the closed-set caveat explicitly.

11. Open items, in priority order

1. **The arbiter run.** Layer-weighted **and** fine-tuned SSL, open-set, one run per language. v1 measured layer-weighted+frozen (2.369 si) and single-layer+fine-tuned (5.404 si) but **never the diagonal that P3's headline claim actually rests on** --- and that claim currently underpins papers/ieee_spl. Named as the highest-value next experiment on 17 August and still unrun as of 10 September.
2. **Seed replication** for the top three systems per language, and for A9. Converts single-run numbers into intervals; without it Stage D's question ("is the ordering stable?") is unanswered.
3. **A7 PLLW on combined_si_ta** --- per-language layer weighting is the direct extension of Finding III-3 for ~39 extra parameters. If the optimal layer differs between Sinhala and Tamil, one shared weight vector is leaving performance on the table.
4. **Complete the T-series** --- T1 Tamil, the no-attention ablation, T2--T4, and T6's span probe (answerable on an already-trained model with no further training).
5. **Fix and re-run A4/A5 on Tamil.** The diagnosis and the one-line fix are recorded; the runs simply never produced a number.
6. **SLCeleb** remains the binding blocker on any Sri Lankan claim. It is the only corpus whose absolute EER would mean what a reader assumes.
7. **Acquire within-speaker, cross-channel Sri Lankan recordings.** This unlocks four of the nine proposals at once and is a clearer data priority than any individual method.
8. **A listening pass on slr52_sinhala** before its Sinhala numbers carry weight. The shortlists exist; no metric can close this.
9. **Run V4 (SE) and V5 (Res2Net) properly.** The Phase II designs were never executed but the rationales are sound and both are cheap under the v1 harness.
10. **Escalate cluster stability.** Four reboots in five days; compute-node-2 needs root to reload its driver (recovering it adds 2 GPUs).
11. **The mel-scale question is untested** since RawNet3 was removed on request --- whether a scale fitted to English and European perceptual data suits Sinhala and Tamil remains open, and is a genuinely publishable question.

12. Assessment

12.1 Achievements of the period

Set out first, because the sections above are organised by chronology and by failure, and the record of what the period actually delivered is otherwise distributed across all twelve of them. Everything listed here is **[M]** --- measured, with an artefact in the repository.

A. Research contributions --- results that are publishable as they stand

#	Contribution	Standing
A1	Speaker information in an SSL encoder falls monotonically with depth, and the conventional last-layer default is the worst available choice. Learned weighting over all 13 states beats every fixed depth, for 13 scalar parameters (Finding III-3)	si 8.247 -> 2.369 held-out; paired bootstrap p = 0.000. The programme's headline result
A2	The last-layer penalty scales with linguistic distance --- 1.04x English, 1.88x Sinhala, 5.23x Tamil. Last-layer SSL practice is an English-centric default whose cost is a function of the target language (Finding III-4)	A mechanism, not just an effect; the English control is what made it visible
A3	Multilingual pretraining pays at matched capacity. mHuBERT-147 beats WavLM-base-plus on both languages at parameter counts matched to within 0.01 %, isolating coverage from capacity	si p = 0.000, ta p = 0.005
A4	Embedding distillation requires an angular loss. MSE on L2-normalised embeddings is unusable --- the normalisation constraint compresses its dynamic range below the classification gradient scale (Finding I-1)	Derived geometrically, confirmed empirically at 36 % relative. Transfers to any normalised-embedding distillation
A5	The optimal distillation weight tracks the student/teacher capacity ratio (Finding I-2)	+4.73 pp recovered by the predicted correction

#	Contribution	Standing
A6	Nested/dense connectivity is domain-specific, not universal --- five measured compatibility criteria, of which spectrogram audio meets none (Finding I-3)	Includes falsifiable predictions for video, ASR and time-series
A7	This low-resource regime is data-limited, not capacity-limited (Finding III-5)	A null with a tight interval: 2.4x the parameters, $p = 0.835$
A8	Genuine cross-session audio is 4--5x harder than read speech for every backbone (Finding III-6) --- and the same-recording confound inflates apparent separability by 42 %, measured directly	The single most important caveat on the whole low-resource SV read-speech literature
A9	Per-language calibration is mandatory (Finding III-1), with the publishable nuance that PLDA scores are the most transportable across languages even where their EER is worse	Cllr inflation up to 5.7x; independently reproduced on hand-crafted features
A10	A controlled null: metadata conditioning adds nothing over score calibration on these corpora, demonstrated by a bit-identical result under metadata shuffling rather than by absent significance (A3/P2)	Stops the next group spending months on the same idea --- a genuine contribution in a literature with almost no published negatives
A11	Methods undefined on the data still train, converge and emit plausible numbers (Finding III-7) --- four of nine literature proposals are undefined on the conditions the benchmark was built from, and none of their papers says so	Generalises well beyond this project

#	Contribution	Standing
A12	The $O(T^2)$ objection does not apply at speaker-verification input lengths (Finding III-8) --- with x4 subsampling the quadratic term is 8.9 % of attention's own cost, and ``efficient'' attention is more expensive	Fitted MAC exponent 1.134, not 2
A13	Transformer value on low-resource languages is in the pretraining, not the architecture --- the same family loses by 0.665 pp from scratch and wins by 1.92 pp pretrained and correctly read, on identical trials	The clean negative the T-series was designed to produce

B. Systems and measured results

Achievement	Figure
First Sinhala/Tamil speaker-verification benchmark of any kind --- protocol, corpora, trial design, metrics	v0 (Jul) then v1 open-set (Aug)
First zero-shot Sinhala EER table across modern architectures --- no Sinhala result exists for any of them anywhere in the literature	4 systems x 2 languages, 12k pairs each
Best Sinhala system (ssl_mhubert_lw), held-out, speaker-disjoint	2.369 % cosine / 2.178 % AS-Norm, minDCF 0.156
Best Tamil system (ssl_mhubert_ecapa)	0.402 % cosine / 0.321 % AS-Norm, minDCF 0.026
Layer-weighted SSL front end vs the strongest mel baseline	46 % better (si), 55 % better (ta)
Best training-free system --- ReDimNet-B6 + AS-Norm, no training, one cohort file	si 2.71 -> 2.07 % , ta 1.48 -> 0.90 %
31k in-language utterances take a 95 M English-pretrained SSL model past the best off-the-shelf supervised model	full FT si 1.69 +/- 0.06 vs ReDimNet-B6 2.71
First positive proposal result , mechanistically coherent with the project's own data audit (A9 sub-centres)	si 4.375 -> 3.962 , -0.413 pp, p = 0.038
Calibration as a free win --- Cllr improved by a median factor of 4.5x at 1,225 parameters, no GPU, no retraining	0.8356 -> 0.1873 across 33 systems
English positive control validates the whole harness end to end	en_full 3.496 % on VoxCeleb1-O
Phase I teacher architecture --- the one Phase I prediction that was met	15.48 -> 9.68 % , 37 % relative

Achievement	Figure
Phase I student inference efficiency	2.04x faster than teacher at 31 % fewer parameters

C. Data assets built

Achievement	Figure
Eight corpora rebuilt into one VoxCeleb-format collection , each self-contained with metadata, lists and a <code>prepare.py</code>	291,061 files converted, zero conversion errors
Tamil speaker pool expanded --- the binding constraint on statistical power, removed	50 -> 706 speakers (802 after QC correction)
Two speaker-identity bugs found and corrected in data this project had built and published --- caught only by embedding the audio, after passing every metadata-level check	SLR127 531 -> 638 speakers, EER 55.31 % -> 1.07 % ; Kathbath 60 -> 49
A seven-layer dataset quality audit with 13 new tools --- integrity, signal quality, label reliability, trial audit, power, calibration, subgroup fairness	Reusable on any future corpus
The channel confound quantified --- corpus-of-origin predicted from the speaker embedding at 91.2 % against 20 % chance	Now a standing caveat on every cross-corpus number
The same-recording confound measured directly on the only corpus that can produce it	d' 3.46 -> 2.44; 42 % inflation
The project's first cross-lingual number --- same speakers, two languages, one embedding	Tamil 1.10 % -> Tamil/English 2.90 % , a 2.6x penalty
Speaker-disjoint splits for every corpus , deterministic, SHA-256 per emitted list, with four corpora held out entirely as generalisation probes	The shipped lists left untouched, so prior results stay reproducible

D. Methodology and tooling

Achievement	Figure
A 54-experiment staged study across six data conditions, run with the trainer unmodified and invoked as a subprocess --- so every previously published number stays on the same code path	46 with held-out evaluation
A paired speaker-clustered bootstrap validated against synthetic ground truth --- EER matches the repository's own <code>tuneThreshold</code> exactly, a known injected difference is detected, a system against itself correctly finds nothing	2.6x tighter than absolute EER

Achievement	Figure
A measured power ceiling that changed the design: effective sample size is bounded by speaker count, not trial count	Trial lists cut 80k -> 20k pairs at no loss of resolving power
Pre-registered predictions for every loss and architecture, written before any run and scored afterwards --- including the two that were falsified and kept rather than edited away	20 predictions scored; ~40 % falsified
An evaluation-integrity rebuild that caught four defects, three of which produced plausible wrong numbers rather than errors, and closed the whole defect class rather than the instances observed	AST-parsed all 82 argparse defaults; all 54 experiments rebuild with zero backbone mismatch
A reusable proposal harness --- shim, argv-copying trainer, data-derived domain guards, isolated scorer --- so any future method supplying a MainModel or LossFunction is trained on the v1 recipe and scored on the v1 trials with one changed factor	The piece most likely to outlive the specific proposals
Exact-reduction controls , asserted rather than asserted-to-be-similar (window attention with $w \geq T-1$ is bit-identical to full; T2's <code>tf0</code> is the stem-only path; T5's four arms share one <code>state_dict</code>)	18 + 13 self-test checks, 0 failures
~40 engineering defects found and fixed , and the defect classes characterised (Finding III-9)	Every Phase III device exists to convert a silent wrong number into a loud failure
A 2.46x faster training platform and a 45x-faster development loop	809 samples/s; 8 s/epoch on the mini-dataset
Cluster capacity and throughput recovered --- idle GPUs found and added, memory-aware scheduling, and the NFS contention diagnosed	3 -> 5 GPUs ; 30--100x throughput recovery by reducing concurrency
No training work lost across four cluster-wide reboots in five days	Per-epoch checkpointing + recorder-resume; verified gap-free, duplicate-free epoch sequences across every boundary
Complete measured evaluation of an external proposal --- all four hand-crafted frameworks implemented and run with controls, inverting the source report's ranking and identifying the one component worth keeping	LSP at 24 bytes, best classical descriptor

E. The achievement that underwrites the rest

The programme identified its own evidence problem, documented it, and rebuilt its methodology around it --- then produced, in the six weeks that followed, more verified results than the preceding thirteen months combined. The audit of 3 July was written by the project about the project; nothing external forced it. That is the capability the year established, and it is the reason every number in sections 6 and 7 can be quoted.

12.2 Overall assessment

What went wrong. Thirteen of sixteen months produced no verified result on the project's actual subject. Phase I optimised a lightweight architecture in a race the field had already settled, and its best number is unreplicated. Phase II produced a report whose experimental claims the repository does not support. The English baseline was blocked by a one-command fix for over five weeks.

What went right, and it is more than it looks. The programme diagnosed its own evidence problem, wrote it down, and rebuilt around it. In the six weeks after the audit it produced: the first Sinhala/Tamil speaker-verification benchmark of any kind; an eight-corpus collection with a full quality audit that found two identity bugs in its own data; a 54-experiment open-set study with paired statistics and a validated power analysis; a headline finding (Finding III-3) that is a genuine contribution to how SSL front ends should be read for low-resource languages, with a mechanism (III-4) explaining why the conventional default is English-centric; a controlled null worth publishing (A3); a first positive proposal result that follows its own data audit (A9); and a cost result that killed four planned runs before they spent GPU-hours (T5).

The through-line. Every one of those six weeks' contributions came from taking a measurement the project had previously assumed. The speaker key was assumed and was wrong. The session structure was assumed and was wrong. The splits were assumed open-set and were 100 % closed. The SSL layer was assumed adequate and was the worst available. The efficient attention was assumed cheaper and was more expensive. The trial count was assumed to buy power and does not. In every case the assumption was plausible, documented, and false --- and in every case one direct measurement settled it.

That is the finding this year actually produced, and it is worth stating as such: **in low-resource speaker verification the dominant error is not a bad model, it is a good number measured on something other than what it claims.**

Sources: research_logs/ 2025-10-26 -- 2026-08-23; 2026-07-03-project-audit-sota-roadmap.md; experiments/analysis/v1-final/ (54 experiments, 46 with held-out evaluation); proposals/A1--A9; transformer_sv/T1--T6; feature_level_testing/; data/datasets_reports/; Research_Progress_Report_Period{2,3}*.md. voxceleb_trainer/ was unmodified by every Phase III experiment, asserted by selftest.py in each strand.